Statistical Data Analysis

Lecture 2

Marc Corstanje

Where Are We?

- Chapter 1: Introduction
- Chapter 2: Summarizing data
  - Still today: discussion of Assignment 1 & Quiz 1
- Chapter 3: Exploring distributions
  - Section 3.1 – The quantile function and location-scale families
  - Section 3.2 – *QQ*-plots
  - Section 3.3 – Symplots
  - Section 3.4 – Two-sample *QQ*-plots
  - Section 3.5 – Goodness-of-fit tests

Distributions

Distribution Functions

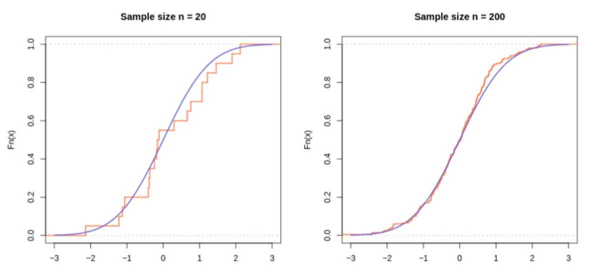
- A (real-valued) random variable is a function $X : \Omega \to \mathbb{R}$
  - Ω is a sample space
- Assumption: sample $x_1, \ldots, x_n$ consists of realizations of i.i.d. random variables $X_1, \ldots, X_n$
  - They all have the same (unknown) probability distribution with distribution function $F(x) = \mathbb{P}(X \leq x)$
  - We want to use the sample to infer about the underlying F

Distribution functions – The ECDF

- The Empirical Cumulative Distribution Function (ECDF) is given by

$$F_n(x) = \frac{\#\{X_i \le x\}}{n} = \sum_{i=1}^{n} 1_{(-\infty,x]}(X_i)$$

- For large n, $F_n(x) \to F(x)$



Goals & Methods

- Goals
  - Underlying distribution $\rightsquigarrow$ statistical model
  - Distributional assumptions $\rightsquigarrow$ statistical tests
- How do we find a distribution that is suitable to the data?
  - Graphical methods
  - Goodness-of-fit tests (next week)
- Which questions can we ask?
  - Is a sample from a symmetric distribution or not?

    $\rightarrow$ histograms, boxplots, symplots
  - Is a sample from a specific distribution or family of distributions?

    $\rightarrow$ QQ-plots, location-scale families, goodness-of-fit tests
  - Are two samples from the same distribution?

    $\rightarrow$ QQ-plots

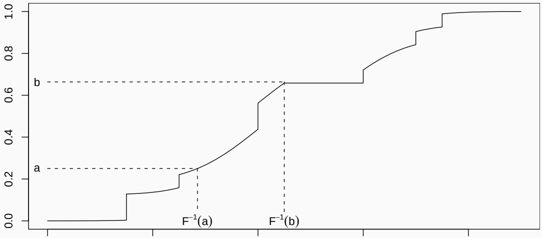
Location-Scale Families

Location-Scale Family

- Take $X \sim F$ and $Y = a + bX$, with $a \in \mathbb{R}$, $b > 0$, so $Y \sim F_{a,b}$
  - $F_{a,b}(x) = F\left(\frac{x-a}{b}\right)$
  - $\mathbb{E}(Y) = a + b \cdot \mathbb{E}(X)$ and $\mathrm{Var}(Y) = b^2 \cdot \mathrm{Var}(X)$
- Location-Scale Family (LSF) w.r.t. F: $\{F_{a,b} : a \in \mathbb{R}, b > 0\}$
- Some examples
  - $X \sim N(0,1)$ and $Y = \mu + \sigma X \sim N(\mu, \sigma^2)$.
  - $Y \sim \mathrm{Exp}(1)$ and $Y \sim \mathrm{Exp}(\lambda)$

Quantile Function

- Quantile function: $F^{-1}(\alpha) = \inf\{x : F(x) \geq \alpha\}, \quad \alpha \in (0,1)$.
  - Inverse of F when F is strictly increasing



- In R: `qnorm`, `qexp`, `qgamma`...
- For an LSF
  - $F_{a,b}^{-1}(\alpha) = a + bF^{-1}(\alpha)$
  - Points $\{(F^{-1}(\alpha), F_{a,b}^{-1}(\alpha)) : \alpha \in (0,1)\}$ are on the line $y = a + bx$

Same LSF?

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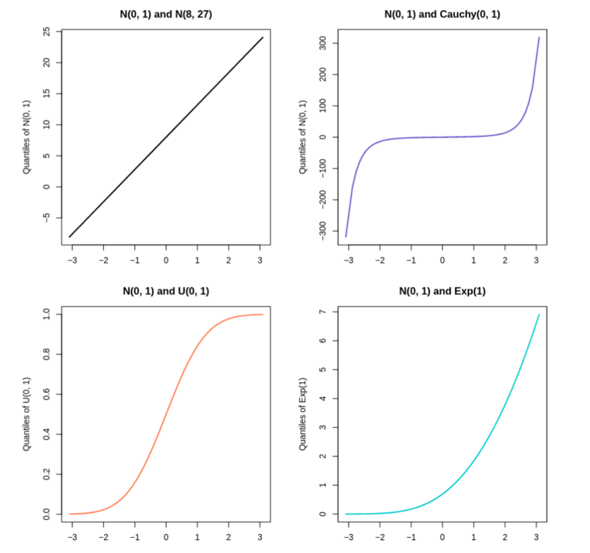
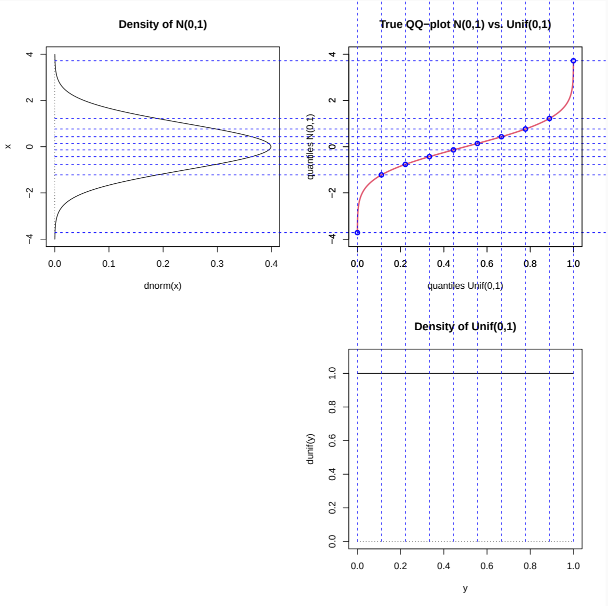


Tail Heaviness



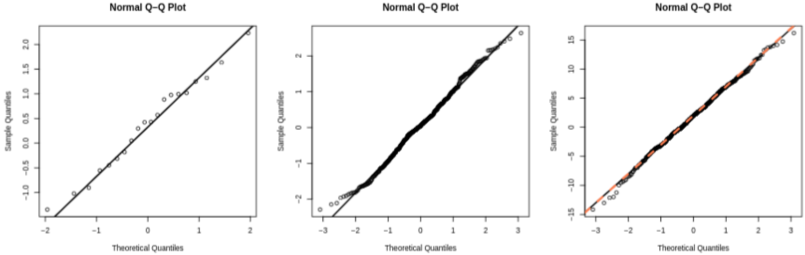
QQ-Plots and Symplots

QQ-Plots

- Idea: Compare theoretical quantile F^{-1} to empirical quantile F_n^{-1}. Could they be from the same LSF?
- Sample quantiles
  - Sample: $x_1, \ldots, x_n$ are real-valued realizations of $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$ continuous
  - For each i, $F_n^{-1}\left(\frac{i}{n+1}\right) = x_{(i)}$
- Points $\left\{\left(F^{-1}\left(\frac{i}{n+1}\right), x_{(i)}\right) : i = 1, \ldots, n\right\}$ should lie on a straight line if F is the underlying distribution of the data
  - If the sample distribution would be *exactly* F, the points would lie on the $y = x$ line
  - If the sample distribution is in the LSF of F, the points lie on a different straight line

QQ-Plots – Samples from $\mathcal{N}(0,1)$ vs. sample from $\mathcal{N}(2,25)$

```r
par(mfrow = c(1, 3))
set.seed(1402)
sample_A <- rnorm(20)
sample_B <- rnorm(500)
sample_C <- rnorm(500, mean = 2, sd = 5)
qqnorm(sample_A)
qqline(sample_A, lwd = 2)
qqnorm(sample_B)
qqline(sample_B, lwd = 2)
qqnorm(sample_C)
qqline(sample_C, lwd = 2)
abline(a = 2, b = 5, col = "coral", lwd = 3, lty = 2)
```



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- Use graphics to investigate the characteristics of the sample distribution
  - Histograms, boxplots and symplots (coming up) can all be used to investigate symmetry, skewness, (relative) heaviness of tails. . .

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- Plot different QQ-plots and choose the one mostly resembling a line.
- Determine the intercept a and the slope b
  - Recall that if $Y = a + bX$, then $\mathbb{E}(Y) = a + b \cdot \mathbb{E}(X)$ and $\mathrm{Var}(Y) = b^2 \cdot \mathrm{Var}(X)$
  - Use $\bar{y}$ and s_Y^2 (sample mean and sample variance of Y), together with $\mathbb{E}(X) := \mu_X$ and $\mathrm{Var}(X) := \sigma_X^2$, to find a and b
  - $b = \sigma_Y / \sigma_X$, estimated by $\hat{b} = s_Y / \sigma_X$
  - $a = \mathbb{E}(Y) - b \cdot \mu_X$, estimated by $\hat{a} = \bar{y} - \hat{b} \cdot \mu_X$
- Also, a so-called two-sample/empirical QQ-plots can be used to check whether two independent samples $x_1, \ldots, x_m$ and $y_1, \ldots, y_n$ originate from distributions in the same LSF

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  - If F is symmetric around θ: $\quad F^{-1}(1-\alpha) - \theta = \theta - F^{-1}(\alpha), \quad \alpha \in (0,1)$
  - Points $\{(\theta - F^{-1}(\alpha),\ F^{-1}(1-\alpha) - \theta) : \alpha \in (0,1)\}$ should lie on the straight line $y = x$
- The symplot of the sample $x_1, \ldots, x_n$ is a plot of the points

$$\left\{(\mathrm{med}(x) - x_{(i)}, x_{(n-i+1)} - \mathrm{med}(x)) : i = 1, \ldots, \left\lfloor \frac{n}{2} \right\rfloor \right\}$$

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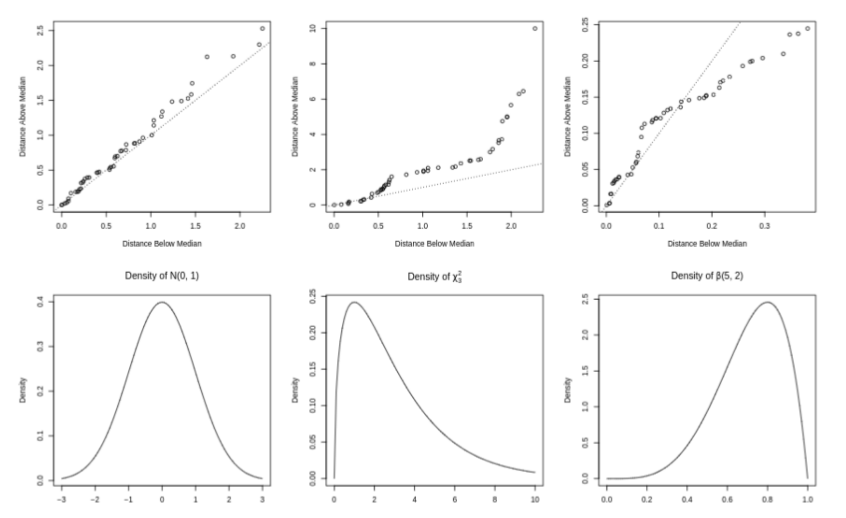
  - The straight line $y = x$ is added to aid in the assessment

Symplots – Sample from $\mathcal{N}(0,1)$ vs. from χ^2_3 vs. from $\beta(5,2)$



Summary and wrap up

Summary

- Location-scale families
  - The distributions of X and Y belong to the same LSF if $Y = a + bX$
  - In that case $F_{a,b}^{-1} = a + bF^{-1}$.
- QQ-plots
  - Plot the quantiles of the data against theoretical quantiles
  - If points are on a straight line, data is from the same LSF
- Symplots
  - Plot the difference of order statistics below median against difference above median
  - If points lie on the line $y = x$, the underlying distribution is symmetric.

Wrap-up

- What now?
  - Start working on assignment 2.
  - Deadline: 24 februray, 23:59.
  - All students still not in pairs by the end of today will be randomly paired.
- Final announcement
  - Tutorial groups 2 and 3 (9:00) are merged into 1 group at 9:00 in NU-3A67